Team 1: 2020-2024 LA Crime Data Analysis

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Abstract

We analyzed the LAPD Public Crime Dataset to examine patterns in crime reports, victim demographics, reporting behavior, and temporal activity in Los Angeles County. Crimes were grouped into time periods (Midday, Evening, Night, Late Night), and chi-squared tests were used to evaluate relationships between variables. Results show significant variation in crime type across time periods and a significant association between victim sex and crime type. Victim age was not significantly related to crime frequency. We also found a significant relationship between crime type and the delay between when a crime occurred and when it was reported. These findings highlight temporal and demographic patterns in crime that may help inform public safety efforts.

Motivation

Los Angeles County experiences persistent crime risk that varies by time and victim characteristics. Understanding these patterns helps reveal when incidents are most likely to occur. This analysis aims to communicate these trends and support future data-driven approaches to improving public safety.

Data

Source: LAPD Crime Data (LAPD CRIME DATA.csv)
Time window: 2020–2023 (2024+ excluded due to incomplete reporting patterns)
Unit: One incident report (DR_NO)
Key fields: DATE.OCC, TIME.OCC, AREA.NAME, Crm.Cd.Desc, Vict.Age, Vict.Sex, Weapon.Used.Cd, Premis.Desc, DATE.RPTD, TIME.RPTD
Cleaning choices: Dropped Crm.Cd.2–4 and Cross.Street due to sparsity and low relevance for our main questions.

Research Questions

- Is these a relationship between the time of day and the type of crime committed in Los Angeles County?
- Is there a relationship between victim demographics (age and sex) and the frequency or type of crime?
- Is there an association between the type of crime and the time between when the crime occurred and when it was reported?

Literature Review

Previous research has examined crime trends in Los Angeles and factors influencing crime distribution. Studies report that while overall crime rates in Los Angeles have declined in recent years, property crimes such as theft remain among the most common incidents. Other research suggests that urban factors, such as dense commercial areas, can create greater opportunities for crime. Building on this work, our study analyzes recent LAPD crime data to explore patterns in crime type, victim demographics, and reporting behavior, with the goal of contributing to a better understanding of public safety trends in Los Angeles County.

Hypotheses (Testable)

Time and Violence

- **H0:** The share of violent crimes does not increase later in the day.
- **H1:** Violent crimes are more common in evening/night hours (18:00–23:59).

Victim Demographics

- **H0:** Victim demographics are not associated with crime type.
- **H1:** Certain crime types show differences by sex or age distribution.

Reporting Time

- **H0:** There is no association between crime type and the time between when a crime occurs and when it is reported.
- **H1:** There is an association between crime type and the time between when a crime occurs and when it is reported.

Results

What the Data Shows (Summary Statistics)

- Vehicle theft is the most common crime; property crime dominates overall
- Victims are primarily young and middle-aged adults (20 - 40 years)
- Temporal trends: crimes are lowest in early morning (4 - 6 AM) and peak midday/evening
- Most crimes are reported quickly, usually the same day
- Gender differences exist, but do not strongly determine victimization patterns

Methods

Exploratory Data Analysis - Analyzed LAPD crime dataset (2020-2024) with 877,327 incidents - Focused on primary crime; removed secondary/tertiary crimes and incomplete 2024 - Examined temporal patterns such as hour of day and reporting delays - Investigated victim demographics, weapon involvement, and crime tpe composition - Visualized patterns using line graphs, box plots, and mosaic plots

Planned Follow-Up Analysis - Apply ANOVA and regression to test relationships between time, area, and crime type - Explore predictive modeling for reporting delays and crime occurrence - Examine correlations between crime and neighborhoods characteristics, land use, or demographics - Analyze weapon-related patterns in relation to crime type and time

Key Findings

- **Time and Violent Crime:** Violent crimes represent a larger share of incidents during evening and night hours compared to midday, supporting the hypothesis that violent crime becomes more common later in the day.
- **Victim Demographics:** Crime types show differences by victim sex, while victim age shows only a weak relationship with crime frequency. Most reported victims fall within the 20–40 age range.
- **Reporting Time:** Crime type is significantly associated with the delay between when a crime occurs and when it is reported, although most incidents are reported the same day or shortly after occurring.

Limitations

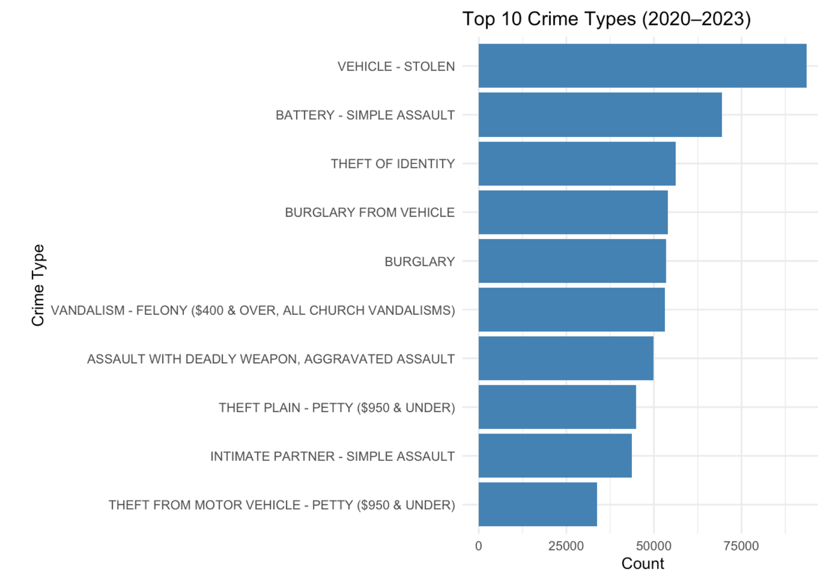
- The dataset only includes reported crimes. Unreported incidents are not captures, which may underestimate the true level of crime
- Some data had to be removed during cleaning. Incomplete 2024 records and variable with many missing values were excluded from the analysis.
- Our analysis is descriptive rather than causal. It identifies patterns in the data, but does not determine why those patterns occur.

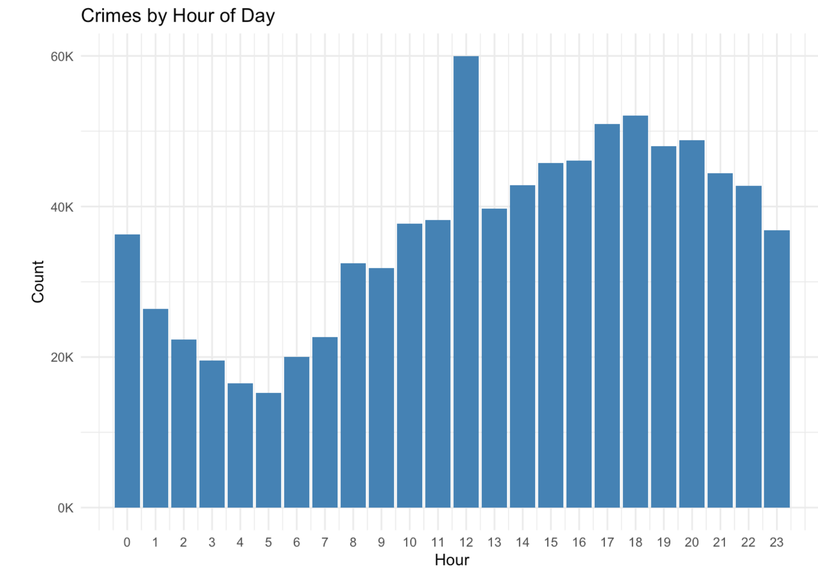
Future Work:

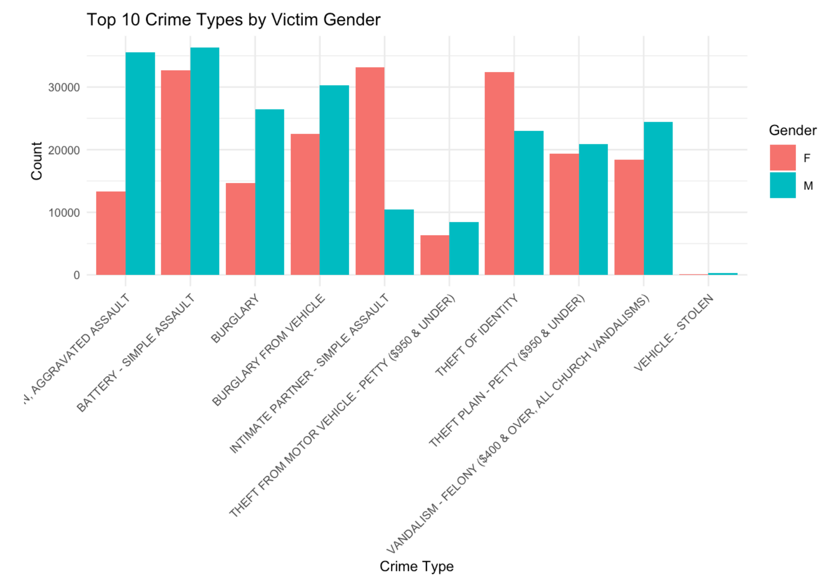
- Apply predictive modeling to examine factor influencing crime occurrence and reporting delays
- Explore relationships between crime and neighborhood characteristics, economic conditions, and land use
- Investigate weapon-related patterns and correlations with crime type or time of day

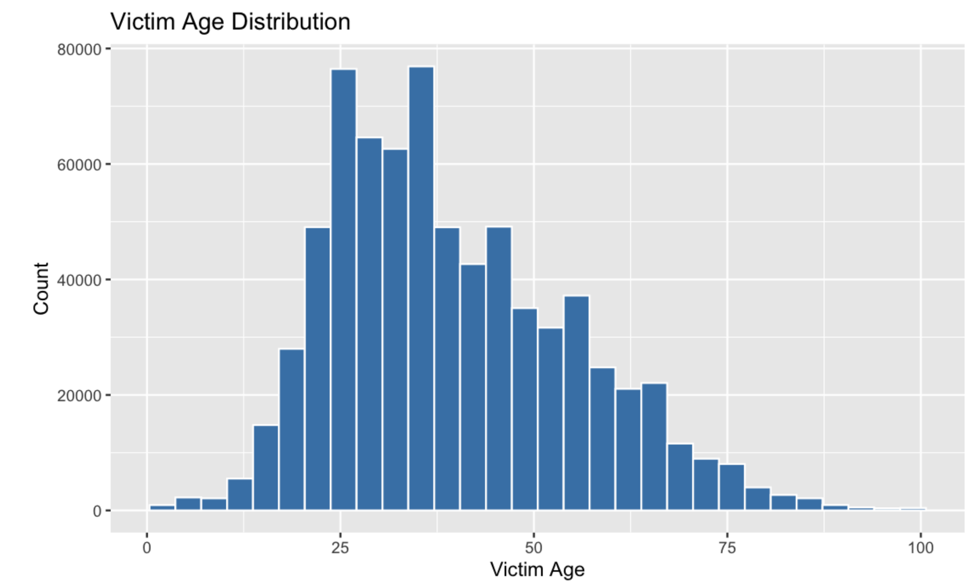
Visuals

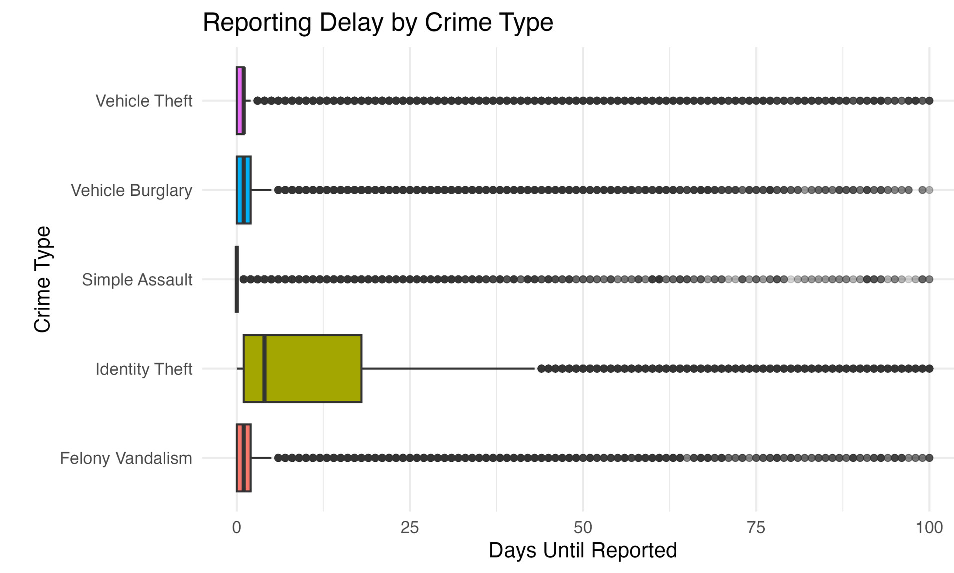
Top Crime Types











References

Newton, Jim. Is crime rising or falling? In Los Angeles, the answer is both, and leaders are struggling to respond. CalMatters (2023).

Anderson, James M., et al. Reducing Crime by Shaping the Built Environment with Zoning: An Empirical Study of Los Angeles. University of Pennsylvania Law Review (2013).